

· AI-POWERED ADAPTIVE LEARNING · PLACEMENT PREP ·

PREPNEXT.

A full-stack, production-deployed platform that builds personalized learning roadmaps and adapts every lesson, hint, and quiz to the individual learner — with a centralized Placement Training Hub and secure user accounts.

Full-Stack

REACT · NODE · MYSQL

Live

PREPNEXTBETA.VERCEL.APP

AI

ADAPTIVE ENGINE

What is PrepNext?

- ▶ **Adaptive learning engine** — models your skill per topic and reshapes lessons, difficulty, and review to fit you.
- ▶ **Placement Training Hub** — one place for DSA, Web, ML, AI, App Dev, Aptitude, Core CS & interviews (10 sections).
- ▶ **DSA practice hub** — 150 curated problems with in-platform solutions, complexity analysis & LeetCode links.
- ▶ **Secure accounts** — email/password auth with cloud-synced, per-user progress.
- ▶ **Audience** — students, job-seekers, career switchers, and professionals prepping for placements & interviews.

How the system fits together

React SPA (TypeScript + Vite)
served as static files on Vercel

▼ **fetch /api/***

Express API as a Vercel serverless function
api/index.js → server/app.js

▼ **Prisma 7 + mariadb adapter**

MySQL database
Railway (production) · local MySQL (development)

One Vercel domain serves both the app and the API. Push to **main** → Vercel builds & deploys automatically (CI/CD).

MySQL · Railway · Vercel

Vercel

Hosting + compute. Serves the static React app and runs the Express API as on-demand **serverless functions** that scale to zero. Auto-deploys on every push to main.

Railway

Managed **MySQL server** in the cloud (where data physically lives). Vercel connects to it over a public proxy URL. Secrets stay in env vars, never in code.

MySQL + Prisma

MySQL is the relational engine; **Prisma 7** (ORM) defines the schema, runs versioned migrations, and turns typed calls like `prisma.user.findUnique()` into SQL.

Login flow — bcrypt + JWT

- ▶ **Sign up** → password is **bcrypt-hashed** and stored in the User table (never plain text).
- ▶ **Log in** → bcrypt compares the input to the stored hash; on success the server signs a **JWT** with a secret key.
- ▶ **Session** → the browser stores the JWT and sends it on protected requests; `/api/auth/me` verifies the signature.
- ▶ **Route protection** → a centralized guard blocks unauthenticated access; a full-screen auth gate captures the intended page and redirects back after login.
- ▶ **Per-user isolation** → all client state is namespaced by user id, so accounts never share data on one browser.

18-table relational schema (Prisma)

- ▶ **User** — accounts + profile (the anchor)
- ▶ **Course** → **Chapter** → **Lesson** — generated courses
- ▶ **Roadmap** — AI learning roadmaps
- ▶ **MasteryEntry** — per-topic skill scores
- ▶ **SRSItem** — spaced-repetition queue
- ▶ **TutorThread** → **TutorMessage** — AI tutor chats
- ▶ **Certificate** — issued certificates
- ▶ **Dsa*** — problem status, bookmarks, attempts
- ▶ **Engagement*** — analytics & interventions
- ▶ **Note** · **Notification**

Normalized to 3NF with primary & foreign keys, unique constraints, indexes, and cascade deletes.
Deep AI payloads stored as JSON columns. Evolved via committed migrations.

The adaptive engine

Mastery scoring

Per-topic EWMA model updates from every lesson, quiz, and solve.

Difficulty targeting

Questions aim at your ~70% success band — the desirable-difficulty sweet spot.

Multi-style lessons

Visual / Code-first / Analogy / Step-by-step, switchable per lesson.

Spaced repetition

SM-2-lite resurfaces weak topics & unsolved problems on the right day.

Engagement watchdog

Tracks focus, scroll & tab-switches; intervenes when you stall.

AI tutor

Socratic guidance & re-explanations without spoiling the answer.

· TECH STACK ·

Built with

Frontend	React 19 · TypeScript · Vite 7 · Tailwind v4 · React Router · framer-motion
Backend	Node 20 · Express 4 (ESM) · serverless functions
Database	MySQL · Prisma 7 ORM · mariadb driver adapter · SQL migrations
Auth	bcrypt (password hashing) · JSON Web Tokens (sessions)
AI	Groq LLM API · strict JSON mode · schema validation + retry
Infra / DevOps	Vercel (hosting + serverless) · Railway (managed MySQL) · Git CI/CD

· THANK YOU ·

Let's connect.

PrepNext — AI-powered adaptive learning & placement prep.

Live demo: prepnexbeta.vercel.app

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